

证明. 若

$$\begin{aligned}\hat{P}^2 &= I \\ \Rightarrow \hat{P}^{2n} &= I, \quad \hat{P}^{2n+1} = \hat{P}\end{aligned}$$

$$\begin{aligned}e^{i\theta\hat{P}} &= \sum_{n=0}^{\infty} \frac{(i\theta\hat{P})^n}{n!} = \sum_{n=0}^{\infty} \frac{(i\theta)^{2n}}{(2n)!} \hat{P}^{2n} + \sum_{n=0}^{\infty} \frac{(i\theta)^{2n+1}}{(2n+1)!} \hat{P}^{2n+1} \\ &= \sum_{n=0}^{\infty} \frac{(i\theta)^{2n}}{(2n)!} I + \sum_{n=0}^{\infty} \frac{(i\theta)^{2n+1}}{(2n+1)!} \hat{P} \\ &= \cos \theta I + i \sin \theta \hat{P}\end{aligned}$$

而 $\hat{P}$ 取:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

均满足 $\hat{P}^2 = I$

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